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Synaptic pruning in Schizophrenia is not *Classical*

Agnieszka Kalinowski [1,*], Alexander E Urban [1,2]

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In Kostkuvi, et al. the authors test the primacy of Classical Complement Pathway (CCP) in mediating reduced synaptic density in SCZ. *In vivo* imaging and postmortem data show that individuals with SCZ have reduced synaptic density.[1] Synaptic density is reduced when C4A gene expression is elevated in animal and *in vitro* cell-based models.[2–4] C4A gene copy number mediates microglial phagocytosis of synaptic material when CCP factors are present *in vitro*.[3] The CCP is known to prune synapses using C1, C2, C3 and C4 proteins.[5,6] Increased CCP activity has been described in other diseases states such as Alzheimer's Disease, Multiple Sclerosis, and Stroke.[7] In those disease states, elevated CCP factors are found in clinical tissue and associated with synaptic reduction and cognitive impairment. High C4A gene risk variants, gene expression and evidence of C4 protein in clinical tissue suggests the CCP may be causing the synaptic density reduction in SCZ. An overactive CCP would provide a very clear treatment target that would potentially preserve

synapses from pathological pruning. However, evidence of increased CCP factors (other than C4) that would be expected to be observed if this pathway were involved in SCZ clinical tissue is lacking.

In this study, Kostkuvi, et al. tested whether the CCP is upregulated in SCZ by measuring the concentration of C1q in brain and cerebrospinal fluid. C1q initiates the CCP and is unique to this pathway. It is also found at elevated levels at synapses in postmortem tissue and cerebrospinal fluid in other disease states. Kostkuvi, et al. used large publicly available databases to compare the gene expression of C1q (A, B and C subunits) and C4A gene expression in SCZ postmortem tissue to controls. They validate that C4A gene expression is upregulated in SCZ samples compared to controls in a substantially larger sample size (approximately ten-fold larger) compared to prior work.[5] However, they find that C1q gene expression is lower in SCZ samples, counter to the expectation that C1q gene expression would be elevated if the CCP were upregulated in SCZ. Likewise, they find lower C1q protein concentration in cerebrospinal fluid in individuals experiencing their First Episode of Psychosis (FEP) compared to controls. Samples from FEP participants are advantageous because they lack the confounds related to chronic disease. Prior work by the same group showed elevated C4A peptide fragment concentration in cerebrospinal fluid in the same clinical cohort.[8] The unexpected combination of low C1q and high C4A differs from other disease states in which CCP factors are elevated, suggesting the involvement of the CCP in these diseases.

To further investigate if the CCP is a core pathological process in SCZ, the authors examine the correlation of gene expression between C1q and C4A. If the CCP is responsible for the reduced synaptic density found in SCZ, we would expect C1q and C4A gene expression (as well as other molecules in the CCP) to be tightly correlated. However, they found diminished correlation in C1q and C4A gene expression in SCZ samples compared to control subjects, suggesting that the CCP is not active in SCZ. The authors also examined the co-expression network of C1q. Prior work showed that C4A gene co-expression networks overlap with Genome-Wide Association Studies (GWAS)[9]; the authors reasoned that examining the co-expression networks for C1q may tell us whether C1q is also related to the expression of SCZ-risk genes and thus more likely to be related to the core pathological process in SCZ. However, Kostkuvi, et al. find that C1q co-expression networks do not overlap with the gene expression of SCZ GWAS risk genes.

The above findings suggest that CCP is not upregulated in SCZ. In fact, C1q gene expression may be downregulated in response to the upregulated C4A gene expression in SCZ. High C4A gene expression is associated with SCZ-risk associated variant haplotypes, generally a greater number of C4A gene copies.[5] CCP factors may be consumed by excessive synaptic pruning driven by high C4A protein expression, and in a negative feedback mechanism, the expression levels of these factors could be downregulated. Alternatively, excessive synaptic reduction in SCZ could be occurring by another mechanism.

Synaptic pruning can occur through other mechanisms that are independent of the CCP.[7] The CCP has been shown to involve/require C4 in neurodevelopment, aging and some diseases. However, astrocytes use MEGF10 and MERTK receptors to signal phagocytosis. Phosphotidylserine, a membrane phospholipid normally found on the intracellular surface of the plasma membrane, triggers microglial phagocytosis when presented on the exterior surface. Additionally, expression levels of complement proteins vary over the course of neurodevelopment, and do not appear to be synchronized among complement proteins in a specific complement pathway. For example, C1q gene expression increases with age. C4 protein concentration peaks in the brain at ages 1–5. Meanwhile, complement inhibitors begin to increase in adolescence. These differences likely also vary in amongst different brain regions.

There is increasing evidence that reducations in synapses are likely a core pathological feature in SCZ. GWAS studies point to a large proportion of risk genes for SCZ are involved in synaptic biology.[10] Experiments in animal models demonstrate how perturbations in neurodevelopment contribute to changes in synapses.[7] However, the key to making a meaningful impact on SCZ treatment will be to figure out if there is an ongoing pathological process that is causing synaptic reduction. Kostkuvi, et al. tests whether a highly active CCP could be destroying useful synapses in SCZ. Unfortunately, they show us that this is unlikely the case.

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